Survey Workbench v2.0

User Manual

February 2026

Version 2.0 – Internal Use Only

Contents

1 Overview

The Survey Workbench is a comprehensive tool for managing participant folders and extracting survey data from questionnaires. Version 2.0 introduces dynamic questionnaire support, allowing unlimited questionnaire types per participant.

1.1 Key Features

- Dynamic questionnaire configuration (unlimited types)
- Automated participant folder generation
- Data extraction from completed questionnaires to Excel
- Configuration save/load system
- Individual questionnaire customization
- Memory-optimized processing

1.2 System Requirements

- Windows 10/11
- Microsoft Excel (for data extraction)
- Sufficient disk space for participant folders

2 Getting Started

When you first launch Survey Workbench v2.0, you'll see the main window with two primary sections:

1. **Generate Participant Folder** – Creates organized folders with questionnaires
2. **Extract Data** – Reads completed questionnaires and appends to masterfile (CSV, XLS, or XLSX)

3 Generate Participant Folder

This section allows you to create standardized participant folders with multiple questionnaire types.

3.1 Step-by-Step Guide

3.1.1 Step 1: Enter Participant ID

Enter a unique identifier for the participant (e.g., "WaS3_001").

- This ID will be used in folder and file names
- Use consistent naming across all participants
- The field cannot be empty

3.1.2 Step 2: Select Target Folder

1. Click **Search** next to "Target folder PATH"
2. Choose where participant folders will be created
3. Example: `C:\Studies\WaS3\Participants`

3.1.3 Step 3: Configure Questionnaire Types

1. Enter the number of different questionnaire types needed
2. Click **Confirm** to create configuration rows

A scrollable area will appear with configuration fields for each questionnaire type.

3.1.4 Step 4: Configure Each Questionnaire

For each questionnaire type, you'll configure three parameters:

Survey Name:

- Enter the questionnaire identifier (e.g., "demographics", "nasa1", "comments")
- This name will be part of the output filename
- Use descriptive, consistent names

Template PATH:

- Click **Search** to select the PDF template file
- Example: `C:\Templates\WaS3_demographics.pdf`
- Template must be a valid PDF file

Number of copies:

- Enter how many copies of this questionnaire to create
- Default is 1
- Use multiple copies for repeated measures (e.g., 8 copies for NASA-TLX)

3.1.5 Step 5: Generate Folder

Click **Generate Participant Folder**. The system will:

1. Create a folder named with the Participant ID
2. Copy all questionnaires with standardized naming
3. Open the folder automatically

3.2 Batch Mode (Generation)

To generate folders for multiple participants at once:

1. Tick **Batch Mode (multiple IDs)** next to the Participant ID field
2. The single-ID field is replaced by a text area; enter IDs one per line or comma-separated
3. Alternatively click **Import List...** to load IDs from a `.txt` or `.csv` file
4. Configure questionnaires as usual and click **Generate Participant Folder**
5. A summary reports how many folders were created and lists any failures

3.3 Template Bundles

A template bundle saves the current questionnaire configuration (names, template paths, copy counts) so it can be reused without re-entering it manually.

3.3.1 Save a Bundle

1. Configure all questionnaire types as required
2. Click **Save Template Bundle** and enter a name when prompted
3. The bundle is stored as a JSON file in the `template_bundles` folder next to the application
4. If a bundle with the same name already exists you will be asked whether to overwrite it

3.3.2 Load a Bundle

1. Click **Load Template Bundle**
2. Select a bundle name from the list
3. All questionnaire rows are populated automatically

3.4 File Naming Convention

- Single copy: `{ParticipantID}_{SurveyName}.pdf`
- Multiple copies: `{ParticipantID}_{SurveyName}1.pdf`, `{ParticipantID}_{SurveyName}2.pdf`, etc.

3.5 Example Output

```
WaS3_001/
├── WaS3_001_demographics.pdf
├── WaS3_001_nasa1.pdf
├── WaS3_001_nasa2.pdf
├── WaS3_001_nasa3.pdf
├── WaS3_001_nasa4.pdf
├── WaS3_001_nasa5.pdf
├── WaS3_001_nasa6.pdf
├── WaS3_001_nasa7.pdf
├── WaS3_001_nasa8.pdf
└── WaS3_001_comments.pdf
```

4 Form Field Naming Convention

Survey Workbench recognises the type of each form field by its name. Every field in a questionnaire form must follow this structure:

`(_)NameX(_N)`

Variables in parentheses are optional and depend on the field type.

4.1 Name

The first part of the field name indicates the **kind** of field:

- `Text` – The field is a free-text input field.
- `Check` – The field is a checkbox.

4.2 X – Field Group Number

`X` is a number that identifies the **question group** the field belongs to. Fields that belong to the same question share the same `X`. For example, the "yes" and "no" checkboxes of a binary question both receive the same group number.

4.3 N – Checkbox Option Number (Check only)

`N` is only used for `Check` fields and identifies the specific checkbox within its group:

- **Binary (yes/no):** Use `1` for *yes* and `0` for *no*.
- **Multiple choice:** Number each option sequentially (1, 2, 3, . . .), where each number is higher than the one before.

4.4 _ – Not-Extracted Prefix

Prefixing a field name with an underscore (`_`) marks the field as **excluded from extraction**. Survey Workbench will skip these fields entirely when processing the questionnaire data. The underscore is placed before the `Name` part of the field name.

4.5 How Field Types Are Extracted

The field name determines how Survey Workbench writes the data into the masterfile:

- **Text field** (`TextX`): The content is written as-is into a single column.
- **Checkbox group** (`CheckX_N`): All options in the group share one output column. The value written is the **N** of whichever option is selected. If nothing is selected the value is empty. Examples: `Check2_0` selected ⇒ `0`; `Check1_2` selected ⇒ `2`; binary pair `Check3_0`/`Check3_1` with `Check3_1` ticked ⇒ `1`.
- **Not-extracted field** (`_NameX(_N)`): The field is ignored and does not appear in the masterfile.

4.6 Naming Examples

Field Name	Meaning
`Text1`	Free-text field, group 1
`_Text2`	Free-text field, group 2 – not extracted
`Check3_1`	Checkbox, group 3, yes-option
`Check3_0`	Checkbox, group 3, no-option
`Check4_1`	Checkbox, group 4, option 1 (multiple choice)
`Check4_2`	Checkbox, group 4, option 2 (multiple choice)
`Check4_3`	Checkbox, group 4, option 3 (multiple choice)
`_Check5_1`	Checkbox, group 5 – not extracted

5 Extract Data

This section reads completed questionnaire data and appends it to a masterfile. The system automatically detects the format (CSV, XLS, or XLSX) from your selected file and uses the appropriate method.

5.1 Prerequisites

- Completed questionnaires must have "_Extract Data.csv" files
- CSV files must be in the participant folder
- A masterfile must exist (.csv, .xls, or .xlsx format)
- The masterfile will be updated with new participant data

5.2 Step-by-Step Guide

5.2.1 Step 1: Select Source Folder

1. Click **Search** next to "Source folder PATH"
2. Choose the folder containing participant folders
3. Example: `C:\Studies\WaS3\Completed`

5.2.2 Step 2: Select Masterfile

1. Click **Search** next to "Masterfile PATH"
2. Choose your masterfile (.csv, .xls, or .xlsx)
3. The system will auto-detect the format and use the appropriate extraction method

5.2.3 Step 3: Enter Participant ID

- Enter the ID of the participant whose data you want to extract
- Must match the folder name exactly
- Case-sensitive

5.2.4 Step 4: Extract Data

Click **Extract Data to Masterfile**. The system will automatically detect your file format and proceed accordingly:

For CSV files (.csv)

1. Find all `*_Extract Data.csv` files in the participant folder, sorted alphabetically
2. Group files by base survey name (trailing digits are stripped from the filename)
3. Flatten all groups and copies into a single ordered row: participant ID first, then each copy's fields sorted alphabetically
4. Write the header row only if the masterfile does not exist yet or is empty
5. Append the participant's values positionally as a new row

For Excel files (.xls or .xlsx)

1. Find all `*_Extract Data.csv` files in the participant folder, sorted alphabetically
2. Group files by base survey name: trailing digits are stripped, so `nasa1`, `nasa2` both belong to the `nasa` group
3. For each survey type, find or create a sheet in the workbook named after that type
4. Check row 1 of the sheet: if it is empty, write the column headers now
5. Find the next empty row starting from row 2 (row 1 is always the header row)
6. Flatten all copies into one row (participant ID first, then each copy's fields in alphabetical order) and write positionally by column index
7. Save as `.xlsx`; if the selected file was `.xls` it is converted and the path shown in the interface is updated

5.3 Choosing Masterfile Format

Use CSV (.csv) when:

- You want a universal, plain-text format
- You plan to import data into R, Python, SPSS, or other analysis tools
- You need maximum compatibility across platforms and software
- You want human-readable, version-control-friendly files
- File size is a concern (CSV is smaller)

Use Excel (.xls or .xlsx) when:

- You want to work directly in Excel with formulas/pivot tables
- You need Excel-specific features (formatting, charts, multiple sheets)
- Your workflow is primarily in Microsoft Excel
- Note: regardless of whether you select a `.xls` or `.xlsx` file, the tool always saves as `.xlsx`; selecting a `.xls` file will convert it automatically and update the path shown in the interface

All formats:

- Append each participant as a new row to the masterfile

- Preserve all previous data in the masterfile
- Column headers are written once on first use; subsequent participants are written by column position
- The tool auto-detects the format – just select your file

5.4 Data Organization

- Participant ID is written in the first column of every data row
- For Excel masterfiles each questionnaire type gets its own sheet, named after the base survey name (e.g. `nasa`, `demographics`)
- All copies of one type are flattened into a single row: participant ID, then first copy's fields, then second copy's fields, and so on
- Fields within each copy are ordered alphabetically (`Check1`, `Check2`, `Text1`, ...)
- Row 1 of each sheet is the header row; data rows start at row 2
- For CSV masterfiles all questionnaire types and copies are flattened into one row per participant
- The `File` column from source CSVs is automatically excluded

5.4.1 Step 5: Preview and Verify

- Before writing, a **preview dialog** shows all field/value pairs that will be extracted; click **Confirm and Extract** to proceed or **Cancel** to abort
- After successful extraction a confirmation message is shown
- Check the masterfile to verify data was written correctly

5.5 Duplicate Detection

Before extraction the tool checks whether the participant ID already exists in the masterfile:

- **CSV:** the `participant_id` column is scanned row by row
- **Excel:** column A (rows 2–1000) of the first sheet is scanned
- In single-participant mode: if a duplicate is found you are asked whether to extract anyway
- In batch mode: duplicates are skipped automatically and listed in the final summary

5.6 Data Completeness Check

Before extraction the tool checks whether all expected `_Extract Data.csv` files are present:

- The expected count is taken from the number of questionnaire types configured in the Generation section
- If the count does not match you are warned and asked whether to extract anyway
- In batch mode: incomplete participants are skipped and listed in the summary

5.7 Batch Mode (Extraction)

To extract data for multiple participants at once:

1. Tick **Batch Mode (multiple IDs)** in the extraction section
2. Enter participant IDs in the text area, one per line or comma-separated; or click **Import List...** to load from a `.txt` or `.csv` file
3. Click **Extract Data to Masterfile**
4. A summary reports successes, duplicates skipped, incomplete participants skipped, and failures
5. Note: the preview dialog is **not** shown in batch mode

5.8 Missing Data Report

Click **Missing Data Report** to scan the entire source folder without extracting any data:

- Each participant subfolder is checked for the presence of `_Extract Data.csv` files
- A report dialog lists which participants are complete and which have missing files
- Useful for quality control before running a batch extraction

6 Configuration Management

Save and load your questionnaire configurations for repeated use.

6.1 Save Configuration

1. Set up your questionnaires in the Generate section
2. Go to **File** → **Save Configuration**
3. Enter a descriptive configuration name
4. Click **Save configuration**

6.1.1 What's Saved

- Target folder path
- Number of questionnaire types
- All questionnaire names, template paths, and copy counts
- Source folder path (for extraction)
- Excel file path (for extraction)

6.1.2 Overwriting Configurations

If a configuration with the same name exists, you'll be asked to confirm:

- Choose **Yes** to replace the old configuration
- Choose **No** to cancel

6.2 Load Configuration

1. Go to **File** → **Load/Delete Configuration**
2. Select a configuration from the dropdown
3. Click **Load configuration**

All saved settings will be restored to the interface.

6.3 Delete Configuration

1. Go to **File** → **Load/Delete Configuration**
2. Select a configuration from the dropdown
3. Click **Delete configuration**
4. Confirm the deletion

7 Troubleshooting

7.1 Common Error Messages

7.1.1 "Please enter a participant ID!"

- Make sure you've entered an ID in the Participant ID field
- The field cannot be empty

7.1.2 "Please select a target folder!"

- Click Search and choose a valid folder
- The folder must exist and be accessible

7.1.3 "Please configure questionnaires!"

- Enter a number in "Number of questionnaire types"
- Click Confirm to create configuration rows

7.1.4 "Participant folder not found"

- Verify the Source folder is correct
- Check that the Participant ID matches the folder name exactly
- Folder names are case-sensitive

7.1.5 "No Extract Data CSV files found"

- Check that questionnaires have been processed
- CSV files must end with "_Extract Data.csv"
- Verify files are in the participant folder

7.1.6 "Folder already exists. Overwrite?"

- Choose **Yes** to replace existing files
- Choose **No** to cancel and preserve existing data

7.2 Excel-Related Issues

- Ensure Excel is installed and accessible
- Close the Excel file if it's already open
- Check file permissions
- Verify the Excel file is not corrupted

7.3 Best Practices

7.3.1 Use Consistent Naming

- Use the same participant ID format across all participants
- Example: WaS3_001, WaS3_002, WaS3_003, etc.
- Avoid special characters in IDs

7.3.2 Organize Templates

- Keep all questionnaire templates in one folder
- Use descriptive filenames
- Maintain backups of template files

7.3.3 Save Configurations

- Create configurations for each study type
- Use descriptive configuration names
- Update configurations when templates change

7.3.4 Backup Data

- Regularly backup your Excel data files
- Keep copies of completed questionnaires
- Use version control for configuration files

7.3.5 Test First

- Test with a sample participant before processing real data
- Verify Excel output is formatted correctly
- Check all questionnaire templates load correctly

8 Advanced Features

8.1 What's This? Help Mode

Survey Workbench has a built-in context-sensitive help system:

- Open **HELP** → **What's This? (Click for Help)** or press **Shift+F1**
- The cursor changes to a question-mark pointer
- Click any control to read a description of its purpose
- The mode exits automatically after clicking a control or pressing **Esc**

8.2 Multiple Questionnaire Copies

When you need repeated measures:

1. Set "Number of copies" to the desired amount
2. Files will be numbered sequentially: `name1.pdf`, `name2.pdf`, etc.
3. Useful for: NASA-TLX repeated measures, multiple trials, time-series data

8.3 Per-Type Excel Sheets

Each questionnaire type automatically receives its own sheet in the workbook:

- The sheet is named after the base survey name (e.g. `nasa`, `demographics`)
- If a sheet with that name does not exist it is created automatically
- Row 1 is always the header row; it is written only once, the first time data is extracted to that sheet
- After the first extraction the user may rename or reorder header cells freely – values are always written by column position, not by matching header names
- All copies of one survey type are written as one row on that sheet

8.4 File Column Filtering

The extraction automatically skips any column named "File" from CSV data to avoid redundant path information.

9 Technical Information

9.1 Version History

9.1.1 v2.0 (February 2026)

- Dynamic questionnaire support (unlimited types)
- Individual configuration per questionnaire
- Improved data extraction algorithm
- Enhanced error handling
- Configuration save/load system

- Memory-optimized processing
- Comprehensive type hints

9.1.2 v1.2 (Previous)

- Fixed questionnaire count
- Basic data extraction
- Simple configuration

9.2 Credits

Version:	2.0
Release Date:	February 2026

9.3 Technology Stack

- Python 3.13.7
- PyQt5 5.15.11 (GUI framework)
- xlwings 0.33.20 (Excel integration)
- PyInstaller 6.14.2 (Executable packaging)

10 Support and Contact

For technical support or feature requests, please refer to your organization's support channels.